

SOLAR Reproducibility Instructions

- We describe the reproducibility instructions for generating the tables and figures included in the main paper. Please refer to the following list to see which section describes the reproducibility of the tables and figures of the main paper, detailed as follows.
- Note that, although reproducible code for the case study is shared, the primary raw data files are not shared in order to comply with data-sharing policies.

Figures

- **Figure 1 (a)-(b)**: concept diagrams, no codes involved.
- **Figure 1 (c)-(g), 2 (a)-(c), 3 (a)-(d), 4 (a)-(d), S1 (a)-(f)**: refer to SOLAR Case Study.

Tables

- **Table 1, 2, S1**: refer to SOLAR Simulation Study.
- **Table 3**: refer to SOLAR Memory scalability study.
- **Table S2-S5**: refer to SOLAR Case Study.

1 Simulation study

1. Navigate to the [Simulation study](#) folder, run [Data_generation.m](#) to generate data for all settings $(n,p) = (500, 10^3), (500, 10^4), (1000, 10^4), (1000, 10^5)$; $q_{\text{true}} = 3, 5$; $(\text{target_SNR_X}, \text{target_SNR_w}) = (0.20, 0.05), (0.01, 0.50), (0.05, 0.20)$.
2. Run [Simulation_PCR.m](#) to fit PCR for all settings.
3. Run [Simulation_PLS.m](#) to fit PLS for all settings.
4. Run [Simulation_study_SPC.m](#) to fit SOLAR for all settings.

5. Run [Simulation_study_SOLAR.m](#) to fit SOLAR for all settings.
6. Run [Summary.m](#) for all considered n , p , q_true , $target_SNR_X$, $target_SNR_w$ to summarize results.
7. Results are saved within [Simulation study/Output_Summary](#) folder for all considered scenarios, and are reported in **Table 1** and **Table S1**.
8. Run [Simulation_SOLAR_ultrahighdim.m](#) (on a ≥ 256 GB RAM, ≥ 10 core HPC cluster node) to fit SOLAR for all sub-scenarios for $(n,p) = (1000, 10^6)$. For this setup, data is generated within the code, and later on used without saving data externally.
9. Run [Simulation_SOLAR_UltraUltrahighdim.m](#) (on a ≥ 400 GB RAM, ≥ 1 core HPC cluster node) to fit SOLAR for all sub-scenarios for $(n,p) = (1000, 10^7)$. For this setup, data is generated within the code, and later on used without saving data externally.
10. Run [Summary.m](#) setting $(n,p) = (1000, 10^6), (1000, 10^7)$ and `method_names = 'SOLAR'` for all considered q_true , $target_SNR_X$, $target_SNR_w$ to summarize results.
11. Results are saved within [Simulation study/Output_Summary](#) folder for all considered scenarios, and are reported in **Table 2**.

2 Memory scalability study

Experiments in this section can be run safely on a 32 GB RAM, 12 CPU cores desktop.

1. Navigate to the [Memory scalability study](#) folder, run [Data_generation_memory_scalability.m](#) to generate data for all settings $(n,p) = (500, 10^6), (1000, 10^6), (2000, 10^6)$.
2. Run [SOLAR.m](#) aforementioned for all settings. Results are saved within [Memory scalability study/Output](#) folder, and are reported in **Table 3**.

3 Case study

Experiments in this section can be run safely on a 32 GB RAM, 12 CPU cores desktop.

1. Navigate to the [Case study/Age methylation data](#) folder, run [1_X_clock_data_epiAge_generation.R](#) to extract data.
2. Run [2_Post_data_extraction_diagnostics.R](#).
3. Run [3_motivation_plots.R](#) to generate the plots motivating the case study, and are saved within [Age methylation data/Motivation plots](#), and are reported in **Figure 1 (c)-(g)**.
4. Run [4_pre_SOLAR_plots_for_aim_1.R](#) to generate diagnostic plots, which are saved within [Age methylation data/EpiAge_diagnostic_plots](#), and are reported in **Figure 2 (a)-(c)**.

5. Navigate to the [Case study](#) folder, run [M1_SOLAR_full_data_Horvath.m](#) to perform and save summary outputs from SOLAR analysis of Horvath residual age acceleration based on full data.
6. Run
 - (a) [M1p1_SOLAR_full_data_Horvath_plots.R](#)
 - (b) [M1p2_SOLAR_full_data_Horvath_enrichment_plot.R](#)
 to generate figures, to be saved within [Output](#) folder, and are shared in **Figure 3 (a)-(d)** and **Figure 4 (a), (b), (d)**.
7. Run [M1boot_SOLAR_full_data_Horvath.m](#) (for parallel bootstrap evaluation one may use HPC cluster nodes).
8. Run [M1p4_SOLAR_full_data_Horvath_bootstrapp_rank_distribution.R](#) to generate **Figure 4 (c)**.
9. Run
 - (a) [M2_SOLAR_full_data_Levine.m](#) (for full data-based Levine analysis)
 - (b) [M2p1_SOLAR_full_data_Levine_plots.R](#)
 to generate figures, to be saved within [Output](#) folder, and are shared in **Figure S1 (a)-(f)**.
10. Run
 - (a) [M1p3_SOLAR_full_data_Horvath_Top_CpG_supp_table.R](#)
 - (b) [M2p3_SOLAR_full_data_Levine_Top_CpG_supp_table.R](#)
 to generate list of top CpGs for Horvath and Levine analyses with other relevant characteristics, to be saved within [Output](#) folder, and are shared in **Table S3, S4, S5**.
11. Run [M4_OOS_PCR_PLS_SOLAR_Horvath.m](#) to perform 80/20 split based out of sample metric study for the Horvath outcome specific SOLAR analysis. Output summary is saved within [Output_OOS](#) folder, and is reported in **Table S2**.